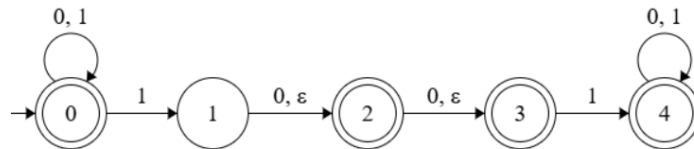


BRAC UNIVERSITY
CSE331 : Automata and Computability
Assignment 2

1. Draw the state diagram of an NFA for the following regular languages:

A. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains } 1001 \text{ or } 11\}$, where $\Sigma = \{0, 1\}$. (use 5 states)



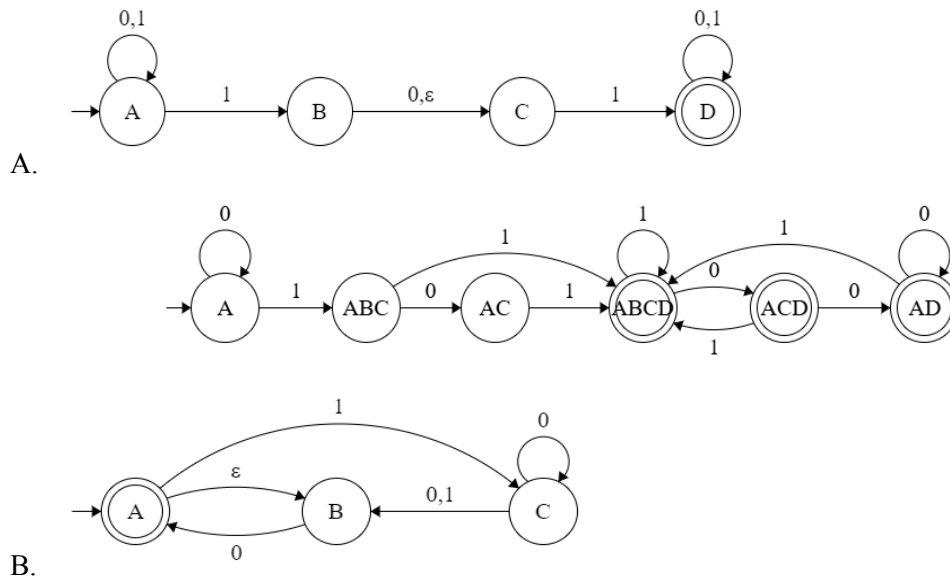
B. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains a } 1 \text{ in the third position from the end}\}$, where $\Sigma = \{0, 1\}$.

See example 1.30 from the book.

C. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{length of } w \text{ is a multiple of } 2 \text{ or } 3\}$, where $\Sigma = \{0, 1\}$.

See example 1.33 from the book.

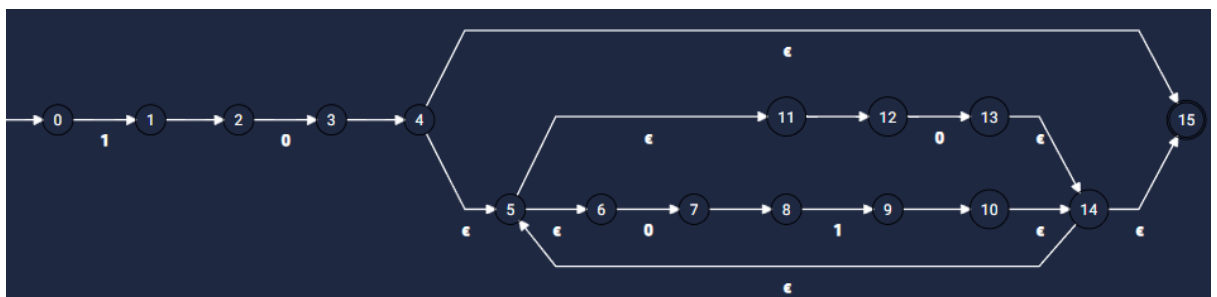
2. Convert the following NFA into DFA:



See example 1.41 from the book.

3. Convert the following RE into NFA:

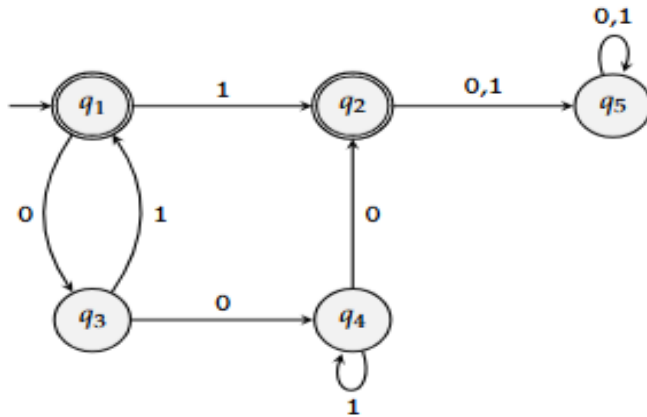
A. $10(01|0)^*$



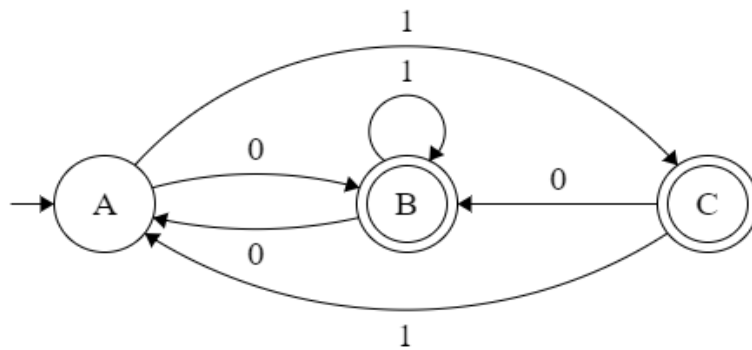
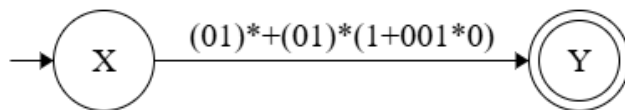
B. $(0 \mid 01^*0)^*01^*0$



4. Convert the following DFA into RE:



A.



B.

See example 1.68 from the book.